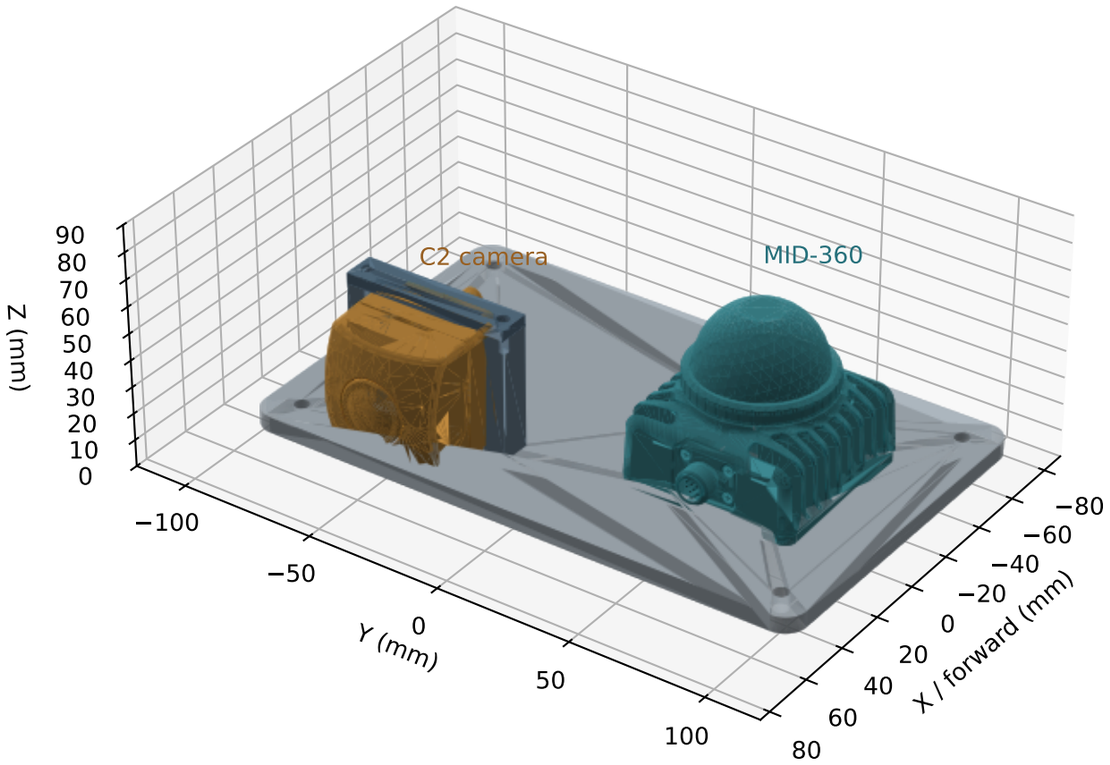
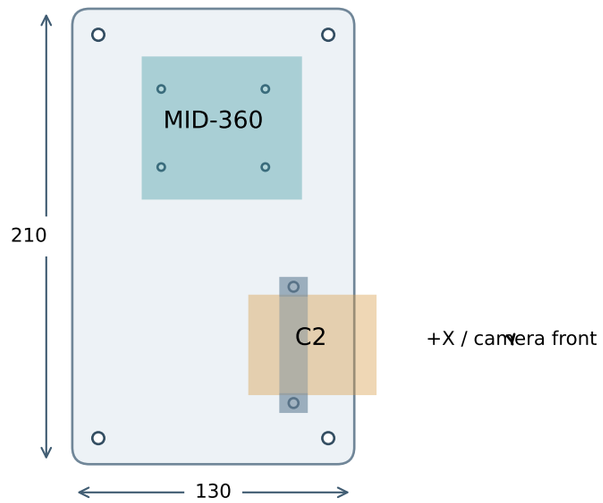
C2-176DEG + MID-360 | Coplanar mounting fixture

USER STEP GEOMETRY / common bottom datum Z=6 mm / mechanical review required

Supplied sensor models + machined parts

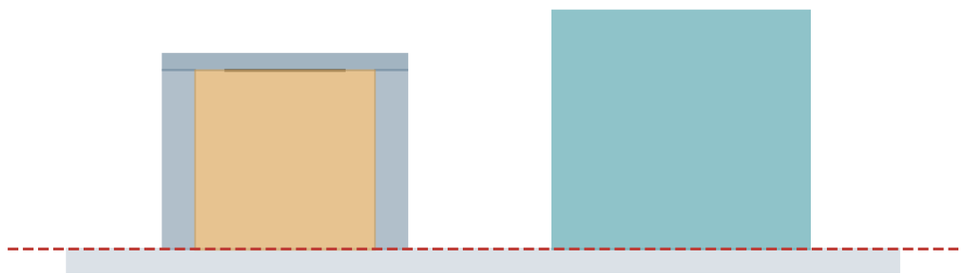


TOP | mm, bounding outlines



FRONT | common underside datum

Camera bottom = LiDAR bottom = Z 6



Base 130 x 210 x 6 mm | A6061-T6 proposed | Camera: top clamp | LiDAR: bottom screws

No bottom pads or height offsets. Thread depth, clamp pressure, cables and FOV need confirmation.

```
FIXTURE | Part dimensions and hole coordinates (mm)

GLOBAL: +X camera forward, +Z up. Plate bottom Z=0, BOTH sensor bottoms Z=6.
BASE: 130(X) x 210(Y) x 6, center (-20,0), R8.
  HOST: dia 5.5 THRU at (-73, -93)
  HOST: dia 5.5 THRU at (33, -93)
  HOST: dia 5.5 THRU at (-73, 93)
  HOST: dia 5.5 THRU at (33, 93)
  LIDAR: dia 3.4 THRU at (-44, 32)
  LIDAR: dia 3.4 THRU at (-44, 68)
  LIDAR: dia 3.4 THRU at (4, 32)
  LIDAR: dia 3.4 THRU at (4, 68)
  POST: dia 4.5 THRU at (17, -76.8)
  POST: dia 4.5 THRU at (17, -23.2)

POSTS (quantity 2): 12(X) x 8(Y) x 45.2 high.
  Centered diameter 3.3 blind drill depth 8 on BOTH ends; tap M4 proposal.
  Threads are not modeled. Top/bottom blind holes are separate.
CAP: 12(X) x 61.6(Y) x 4; 2 x dia 4.5, pitch 53.6.
  Global center (17,-50); underside Z=51.2.
TOP PAD: 8 x 30, compressed thickness 0.2; initial 0.3 proposed.
  Camera top Z=51.0. No pad underneath either sensor.

FASTENERS (proposals; verify lengths and threads):
  LiDAR: 4 x M3; length = 6 mm plate + manufacturer approved insertion.
  Clamp: 4 x M4 (2 top, 2 bottom); choose <8 mm thread engagement.
  Host: 4 x M5; host-dependent length, >=5 mm underside head clearance.

MODEL EVIDENCE:
  C2 source optical +Y -> global +X; bottom Z=-22.5 -> global Z=6.
  MID source upright +Y -> global +Z; bottom Y=-25.9171 -> global Z=6.
  MID holes source X=+/-24, Z=+/-18, 48 x 36 pitch.
  R1.5 entrances and R1.25 inner bores suggest M3; VERIFY actual threads.

REVIEW: cable plugs/bends, clamp pressure, friction, thread depth, FOV, loads.
Plate holes are through clearance holes. Posts require tapping after drilling.
Deburr edges. Mounting face flatness 0.05 and Ra 3.2 proposed; validate needs.
Estimated metal mass: 0.469 kg.
Individual STEP files + coordinate tables define geometry; review before machining.
```